

Muhammad Alyan Asmat

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EDUCATION

University Of Houston

Bachelor of Science, Computer Science

Houston, TX

May 2026

EXPERIENCE

Lead Instructor

Coder Kids

March 2023 – July 2023

Houston, TX

- Designed and delivered programming curriculum to classes of up to 20 students aged 5-10, adapting technical concepts in Scratch and Python across varying skill levels and learning styles
- Led hands-on game development projects using pygame, breaking down control flow and syntax into structured exercises that progressed students from zero to functional programs
- Mentored students through debugging and problem-solving challenges, building independent thinking habits rather than providing direct answers

PROJECTS

One Piece Codenames | *React, TypeScript, Node/Express, Socket.io*

May 2026

- Built a real-time multiplayer word game supporting concurrent players across separate devices, engineered around a server-authoritative architecture to act as the single source of truth for game state.
- Solved the hidden information problem by implementing server-side role-based state filtering, ensuring spymasters and operatives receive different views of the same game without client-side data leaking.
- Curated and structured a custom JSON word list of 220+ One Piece terms organized by category (characters, locations, devil fruits, and more) to replace the standard word set.
- Deployed full-stack application to Render with persistent room management and real-time synchronization across all connected clients.

GuessPKMN | *React, TypeScript, TailwindCSS, Firebase*

September 2025

- Built a full-stack Pokémon guessing game by architecting a custom backend service that normalizes and re-exposes PokeAPI data, decoupling the client from an unstable external dependency.
- Solved a data freshness and performance problem by designing a rolling 50-entity randomization cache with generation-based filtering to enforce uniqueness constraints without repeated API calls.
- Implemented Firebase authentication and a real-time database schema to persist and update user lifetime high scores across sessions.

Cardio Medical | *JavaScript, HTML, CSS, Supabase*

May 2025

- Built a responsive website for a 10-person student medical organization, coordinating requirements gathering and iterative design feedback across the full team.
- Implemented a Supabase backend with user authentication and a real-time blog system, managing secure sessions and live data synchronization.
- Implemented Supabase backend providing user authentication and database management for a blog system with real-time data synchronization and secure user sessions.

Online Portfolio | *React, JavaScript, HTML, CSS*

August 2025

- Built a responsive personal portfolio with reusable React component architecture, optimized for cross-device rendering and interactive UI behavior.
- Configured a CI/CD pipeline using GitHub Pages with custom domain routing through Cloudflare DNS for automated deployments and secure content delivery.
- Deployed portfolio site receiving 1,550+ unique visitors over 30 days.

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, HTML/CSS

Frameworks/Libraries: React.js, Next.js, Node, Express.js, TailwindCSS, Jest, Socket.io

Tools/Technologies: Git, GitHub, MySQL, Postman, Cloudflare, VSCode, Postgres, Figma, Docker, Render, Supabase, Firebase